

Assignment-12

Part-A

1 Find the integrating factor $\frac{dy}{dx} + \frac{3}{x}y = x^2$

$$P = \frac{3}{x}$$

$$IF = e^{\int P dx}$$

$$= e^{\int \frac{3}{x} dx} = e^{3 \ln x} = x^3 \quad \underline{\underline{\text{Ans}}}$$

2 The eqn $\frac{dy}{dx} = \frac{x+y}{x}$ can be solved by separation of variables/exact methods/substitution ($y = vx$)/Bernoulli.

$$\frac{dy}{dx} = 1 + \frac{y}{x}$$

Since RHS is fn of y/x , it is a homogeneous eqn hence we substitution

$$y = vx$$

Substitution ($y = vx$)

3 Check whether $(2xy + 3)dx + (x^2 + 4y)dy = 0$ is exact.

Given $M = 2xy + 3$
 $N = x^2 + 4y$

$$\frac{\partial M}{\partial y} = 2x$$

$$\frac{\partial N}{\partial x} = 2x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Yes, the eqn is exact.



4 Find order and degree of $\left(\frac{dy}{dx}\right)^3 + \left(\frac{dy}{dx}\right)^2 = x$

order = 1 ; degree = 3.

5 Solve : $\left(\frac{dy}{dx}\right)^2 - 4\frac{dy}{dx} + 3 = 0$

$$p = \frac{dy}{dx}$$

$$p^2 - 4p + 3 = 0$$

$$(p-1)(p-3) = 0$$

$$p = 1 \text{ or } 3$$

Cases : ① $\frac{dy}{dx} = 1 \rightarrow \text{Integrate}$

$$y = x + C_1$$

② $\frac{dy}{dx} = 3 \rightarrow \text{Integrate}$

$$y = 3x + C_2$$

Part - B

1 Solve : $3x(1-x^2)y^2 \frac{dy}{dx} + (2x^2-1)y^3 = ax^3$

divide by y^2

$$\frac{3x(1-x^2)}{y^2} \frac{dy}{dx} + (2x^2-1)y = \frac{ax^3}{y^2}$$

This is Bernoulli's Eqn

$$\text{let } v = y^2$$

$$\frac{dv}{dx} = 2y^2 \frac{dy}{dx}$$

$$x(1-x^2) \frac{dv}{dx} + (2x^2-1)v = ax^3$$

$$\frac{dv}{dx} + \frac{2x^2-1}{x(1-x^2)} v = \frac{ax^2}{1-x^2}$$

Integrating factor : $e^{\int \frac{2x^2-1}{x(1-x^2)} dx} \Rightarrow \text{IF} = \frac{x}{\sqrt{1-x^2}}$

$$\text{Thus, } \frac{v \cdot x}{\sqrt{1-x^2}} = \int \frac{ax^3}{(1-x^2)^{3/2}} dx + C$$

substituting $v = y^2$, final soln.

$$\frac{xy^2}{\sqrt{1-x^2}} = a\sqrt{1-x^2} + C$$

$$\frac{2}{2} (1+y^2) + (x - e^{\tan^{-1}y}) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-1+y^2}{x - e^{\tan^{-1}y}}$$

$$\frac{dx}{dy} = \frac{x - e^{\tan^{-1}y}}{1+y^2}$$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{e^{\tan^{-1}y}}{1+y^2}$$

$$IF = e^{\int p(x) dx} \Rightarrow \int \frac{1}{1+y^2} dy = e^{\tan^{-1}y}$$

Integrate :

$$x(IF) = \int (Q \cdot IF) dy$$

$$= \int \frac{e^{\tan^{-1}y}}{1+y^2} \cdot e^{\tan^{-1}y} dy = \int \frac{e^{2\tan^{-1}y}}{1+y^2} dy$$

$$t = \tan^{-1}y$$

$$dt = \frac{dy}{1+y^2}$$

$$xe^t = \int e^{2t} dt + C$$

$$xe^t = \frac{1}{2} e^{2t} + C$$

$$xe^{\tan^{-1}y} = \frac{1}{2} e^{2\tan^{-1}y} + C$$

3. solve : $4yp^2 + 2px - y = 0$

$$p = \frac{dy}{dx}$$

Given $2px = y - 4yp^2$

$$x = \frac{y(1-4p^2)}{2p}$$

$y = x f(p)$. This is Lagrange Form.

$$p = \frac{-2p}{4p^2-1} + x \frac{d}{dp} \left(\frac{-2p}{4p^2-1} \right) \frac{dp}{dx}$$

Diff $\frac{d}{dp} \left(\frac{-2p}{4p^2-1} \right) = \frac{2(4p^2+1)}{(4p^2-1)^2}$

After simplification

$$\frac{dx}{dp} = \frac{-2x(4p^2+1)}{p(4p^2-1)}$$

Seperate variable $\frac{dx}{x} = \frac{-2(4p^2+1)}{p(4p^2-1)} dp$

Integrating: $\ln x = \ln \left(\frac{4p^2-1}{p^2} \right) + \ln C$

$$x = \frac{4p^2-1}{p^2} + C$$

Now,

$$y = \frac{-2px}{4p^2-1}$$

→ substitute x

$$y = \frac{-2C}{p}$$

Part - C

$\frac{1}{2}$ Solve $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$

let $M = 3x^2y^4 + 2xy$; $N = 2x^3y^3 - x^2$

$$\frac{\partial M}{\partial y} = 12x^2y^3 + 2x$$

$$\frac{\partial N}{\partial x} = 6x^2y^3 - 2x$$

Not exact

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

use Integrating Factor: $IF = \frac{1}{y^2}$

Multiplying throughout

$$\left(3yx^2 + \frac{2x}{y}\right) dx + \left(2x^3y - \frac{x^2}{y^2}\right) dy = 0$$

$$M' = 3x^2y^2 + \frac{2x}{y} \quad ; \quad N' = 2x^3y - \frac{x^2}{y^2}$$

$$\frac{\partial N'}{\partial y} = 6x^3y - \frac{2x}{y^3} \quad ; \quad \frac{\partial N'}{\partial x} = 6x^2y - \frac{2x}{y^2} \quad \left(\frac{\partial N'}{\partial x} = \frac{\partial M'}{\partial y}\right)$$

is exact

Now integrate

$$y = \int (3x^2y^2 + \frac{2x}{y}) dx + \int 0 dy$$

$$= \boxed{x^3y^3 + \frac{x^2}{y} = C}$$

$$\approx (x^2y^2 + xy + 1)y dx + (x^2y^2 - xy + 1)x dy = 0$$

let $M = y(x^2y^2 + xy + 1) \quad ; \quad N = x(x^2y^2 - xy + 1)$

let Find the Integrating Factor:

$$IF = \frac{1}{M_x - N_y}$$

$$M_x = x^3 y^3 + x^2 y^2 + xy ; N_y = x^3 y^3 - x^2 y^2 + xy$$

$$M_x - N_y = 2x^2 y^2$$

$$IF = \frac{1}{2x^2 y^2}$$

Multiply IF in original eqn

$$\frac{(x^2 y^2 + xy + 1)y}{2x^2 y^2} dx + \frac{(x^2 y^2 - xy + 1)dy}{2x^2 y^2} = 0$$

$$\left(y + \frac{1}{x} + \frac{1}{x^2 y}\right) dx + \left(x - \frac{1}{y} + \frac{1}{x^2}\right) dy = 0$$

$$y = \int \left(y + \frac{1}{x} + \frac{1}{x^2 y}\right) dx - \int \frac{1}{y} dy = C$$

$$xy + \ln|x| - \ln|y| - \frac{1}{xy} = C$$

$$\boxed{xy + \ln\left|\frac{x}{y}\right| - \frac{1}{xy} = C}$$

3. Solve (i) $\sin px \cos y = \cos px \sin y + P$

$$\sin(px) \cos y - \cos(px) \sin y = P$$

$$\boxed{\sin(A - B) = \text{Identity}}$$

$$\sin(px - y) = P$$

$$px - y = \sin^{-1} P$$

$$y = -\sin^{-1} P + px$$

This is Clairaut's Eqn from $y = px + f(p)$

The general soln for any Eqn of the form
 $y = px + f(p)$ is obtain by replacing
 p with C .

$$\boxed{y = cx - \sin^{-1}(c)}$$

(ii) $x - yp = ap^2$

$$p = \frac{dy}{dx}$$

$$yp = x - ap^2 \Rightarrow y = \frac{x}{p} - ap$$

Diff

$$y' = p = \frac{d}{dx} \left(\frac{x}{p} - ap \right)$$

$$p = \frac{p - x \frac{dp}{dx}}{p^2} - a \frac{dp}{dx}$$

$$p = \frac{1}{p} - \frac{x}{p^2} \frac{dp}{dx} - a \frac{dp}{dx}$$

Multiply the entire Eqn by p^2

$$p^3 = p - x \frac{dp}{dx} - ap^2 \frac{dp}{dx}$$

$$p^3 - p + (x + ap^2) \frac{dp}{dx} = 0$$

$$p(p^2 - 1) + (x + ap^2) \frac{dp}{dx} = 0$$

one part of soln

$$\frac{dp}{dx} = 0 \Rightarrow p = c \text{ (a constant)}$$

General soln: $x - yc = ac^2$

$$\boxed{y = \frac{x}{c} - ac}$$

~~Sub~~ 18/15/20 Substitute $p = c$